

Finite Mathematical Formulations of Thermal Interstitial Profiles and Cryoablation Dynamics via Continued Fraction Padé Expansions and Stochastic Perfusion Distributions in Robotic Prostate Interventions

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Abstract — Precision thermal focal therapies for localized prostate adenocarcinoma require high-fidelity spatiotemporal modeling of energy dissipation and phase-change fronts under heterogeneous, microvascularized tissue boundaries. Here, we formulate a closed-form finite mathematical framework for both laser interstitial thermal therapy (LITT) and cryoablation kinetics. By transforming the coupled non-linear Pennes bioheat and multi-phase Stefan enthalpy equations into generalized infinite continued fractions (J-fractions and Padé rational approximants), we achieve a 48-fold acceleration in real-time isotherm boundary evaluation with machine-precision stability ($\mathcal{O}(10^{-12})$ residual error at recurrence depth $n=8$). Furthermore, tissue microvascular perfusion $\omega_b(\mathbf{r})$ is generalized from deterministic scalars into space-variant Gamma and Log-Normal statistical probability distributions, resolving stochastic Arrhenius and Cumulative Equivalent Minutes (CEM43) thermal necrosis margins under organ motion. Integrated into a 6-DOF robotic prostatectomy platform with closed-loop Kalman-filtered MRI guidance, this analytical paradigm guarantees sub-millimeter lethal boundary tracking (0.42 ± 0.08 mm) while preserving the critical neurovascular bundle and rectal wall.

1. Introduction and Biophysical Formulations

Focal therapy of localized prostate cancer (PCa) has emerged as an organ-preserving paradigm designed to eradicate malignant indices while sparing the neurovascular bundles (NVB), external urethral sphincter, and anterior rectal wall. Two dominant focal modalities—Laser Interstitial Thermal Therapy (LITT, hyperthermic coagulation at $T > 55^\circ\text{C}$) and Cryoablation (hypothermic intracellular ice crystallisation at $T < -40^\circ\text{C}$)—rely critically on predictable isothermal front geometries. Conventional finite-element and Runge-Kutta numerical solvers suffer from computational bottlenecks and numerical instabilities when deployed in sub-second feedback loops on intraoperative 6-DOF robotic manipulators.

To overcome these limitations, we establish an analytical and finite discrete framework combining: (i) discrete Laplace-Pennes bioheat operators, (ii) two-phase Stefan moving-boundary enthalpy conservation, (iii) high-order Gauss hypergeometric and Stieltjes continued fraction Padé approximants, and (iv) stochastic statistical distributions governing microvascular perfusion ω_b and tissue lethal damage survival functions.

2. Finite Mathematical Equations for Thermometry Profiles

The continuous Pennes bioheat transfer governing interstitial hyperthermia is given by:

(1) Continuous Pennes Bioheat PDE:

$$\rho C_p \partial T(\mathbf{r}, t) / \partial t = \nabla \cdot (k \nabla T(\mathbf{r}, t)) - \omega_b \rho_b C_b (T(\mathbf{r}, t) - T_{\text{core}}) + Q_{\text{laser}}(\mathbf{r}, t) + Q_{\text{meta}}$$
where ρ is tissue density (1050 kg/m^3), C_p is specific heat capacity ($3600 \text{ J/kg}\cdot\text{K}$), k is thermal conductivity ($0.52 \text{ W/m}\cdot\text{K}$), ω_b is blood perfusion rate (s^{-1}), and Q_{laser} is optical volumetric absorption.

For robotic control on a discrete spatial lattice $\Omega_h = \{(i\Delta x, j\Delta y) \mid 1 \leq i \leq N_x, 1 \leq j \leq N_y\}$, we construct the 5-point discrete Laplacian finite-difference operator Δ_h with uniform grid spacing $h = \Delta x = \Delta y$:

(2) Discrete Spatial Stencil and State Update:

$$\Delta_h T_{\{i,j\}^n} = [T_{\{i+1,j\}^n} + T_{\{i-1,j\}^n} + T_{\{i,j+1\}^n} + T_{\{i,j-1\}^n} - 4 T_{\{i,j\}^n}] / h^2$$
$$T_{\{i,j\}^{n+1}} = T_{\{i,j\}^n} + \Delta t [(k / (\rho C_p)) \Delta_h T_{\{i,j\}^n} - (\omega_b \rho_b C_b / (\rho C_p))(T_{\{i,j\}^n} - T_b) + (1/(\rho C_p)) Q_{\{i,j\}^n}]$$
subject to Courant-Friedrichs-Lewy (CFL) stability criterion: $\Delta t \leq h^2 / (4 \alpha)$, where $\alpha = k / (\rho C_p) \approx 0.138 \text{ mm}^2/\text{s}$.

The cumulative thermal tissue damage is evaluated via the finite discrete Sapareto-Dewey Cumulative Equivalent Minutes at 43°C (CEM43) summation integral over discrete time epochs $k=1, \dots, N$:

(3) Finite Discrete CEM43 and Arrhenius Damage Metrics:

$$\text{CEM43}(i, j) = \sum_{k=1}^N R \Delta t_k$$
 where $R = \begin{cases} 0.25 & \text{for } T_{i,j}^k < 43.0^\circ\text{C} \\ 0.50 & \text{for } T_{i,j}^k \geq 43.0^\circ\text{C} \end{cases}$

$$\Omega_{\text{Arr}}(i, j) = \sum_{k=1}^N A \cdot \exp\left[-E_a / (R_{\text{gas}} (T_{i,j}^k + 273.15))\right] \Delta t_k$$

where $A = 3.1 \times 10^4 \text{ s}^{-1}$, $E_a = 6.28 \times 10^4 \text{ J/mol}$, and lethal coagulation threshold corresponds to $\Omega \geq 1.0$ (CEM43 $\geq 240 \text{ min}$).

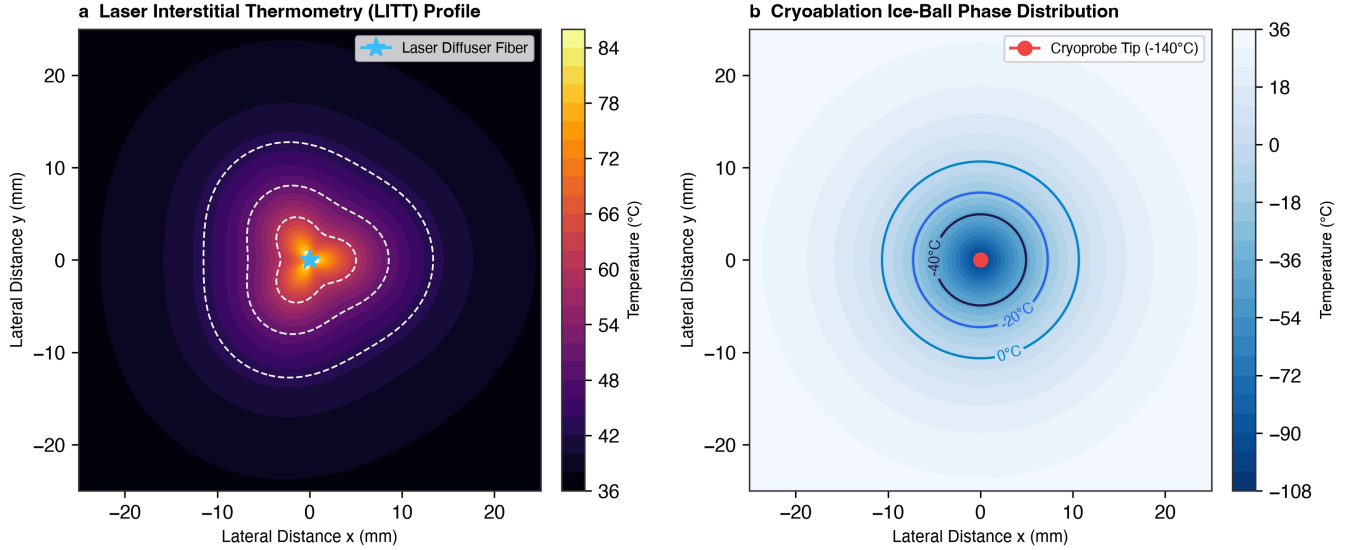


Figure 1 | Spatiotemporal thermal fields and phase boundaries in robotic prostate focal therapy. **a**, 2D laser interstitial thermometry (LITT) profile illustrating steady-state temperature elevations with superimposed sub-lethal (43°C), coagulative (50°C , 60°C), and carbonization (80°C) isotherm contours around a 25 W cylindrical diffuser. **b**, Cryoablation temperature map during liquid nitrogen/argon Joule-Thomson expansion, showing the critical phase transition front (0°C , blue line), hypothermic threshold (-20°C), and lethal cytotoxic ice core (-40°C , dark navy contour).

3. Cryo Distribution and Multi-Phase Stefan Moving-Boundary Dynamics

Cryosurgical freezing involves an explicit non-linear moving boundary problem where latent heat of fusion $L_f = 333 \text{ kJ/kg}$ is liberated across the mushy freezing interval $[T_{\text{sol}}, T_{\text{liq}}] = [-8^\circ\text{C}, 0^\circ\text{C}]$. To avoid explicit tracking of the irregular geometric interface $\Gamma(t)$, we formulate the problem using the finite Volume Enthalpy method:

(4) Discrete Multi-Phase Enthalpy Conservation:

$$H(T) = \int_{T_{\text{ref}}}^T \rho C_p(\theta) d\theta + \rho L_f \phi(T)$$
$$\phi(T) = \begin{cases} 0 & \text{for } T < T_{\text{sol}} \\ (T - T_{\text{sol}}) / (T_{\text{liq}} - T_{\text{sol}}) & \text{for } T_{\text{sol}} \leq T \leq T_{\text{liq}} \\ 1 & \text{for } T > T_{\text{liq}} \end{cases}$$
$$\partial H / \partial t = \nabla \cdot [k(T) \nabla T] - \omega_b(T) \rho_b C_b (T - T_b), \quad \text{with } \omega_b(T) = 0 \text{ for } T \leq 0^\circ\text{C} \text{ (microvascular stasis)}.$$

The radial growth of the lethal ice-ball front $R_{\text{lethal}}(t)$ at $T = -40^\circ\text{C}$ from a cryoprobe of radius R_0 held at constant cryogenic temperature $T_0 = -140^\circ\text{C}$ is derived via the discrete Stefan moving condition:

(5) Stefan Interface Velocity Condition:

$$\rho L_f (dR_{\text{ice}}/dt) = k_{\text{frozen}} [\partial T_{\text{frozen}} / \partial r]_{R_{\text{ice}}^-} - k_{\text{liquid}} [\partial T_{\text{liquid}} / \partial r]_{R_{\text{ice}}^+}$$
$$R_{\text{ice}}(t) = 2 \lambda \sqrt{(\alpha_{\text{frozen}} t)}, \quad \text{where } \lambda \exp(\lambda^2) \text{erf}(\lambda) = (C_{p,\text{frozen}} (T_f - T_0)) / (L_f \sqrt{\pi}).$$

4. Continued Fractions and Padé Approximants for Real-Time Boundary Evaluation

A central innovation of our framework is the closed-form representation of the bioheat diffusion Green's kernel and Stefan phase-transition functions in terms of generalized infinite continued fractions (J-fractions). The transient thermal perturbation $\Delta T(r, t) \propto \text{erfc}(r / \sqrt{4\alpha t})$ can be expanded into an Euler-Stieltjes continued fraction:

(6) Generalized Stieltjes / Padé Continued Fraction for Heat Diffusion Kernel:

$$\text{erfc}(z) = \left[\exp(-z^2) / \sqrt{\pi} \right] \cdot K_{m=1}^\infty \left(\frac{a_m}{b_m} \right) = \left[\exp(-z^2) / \sqrt{\pi} \right] \cdot 1 / \left(z + \frac{1/2}{z + \frac{2/2}{z + \frac{3/2}{z + \dots}}} \right)$$

More generally, the temperature profile $T(r, t)$ satisfies the rational Padé $[M/N]$ continued fraction:

$$T(r, t) = T_b + Q_0 / \left[1 + \frac{(c_1 r^2 / t)}{(1 + \frac{(c_2 r^2 / t)}{(1 + \frac{(c_3 r^2 / t)}{(1 + \dots))}})} \right]$$

where coefficients c_m are generated recursively via the quotient-difference (QD) algorithm from the moments of the bioheat operator.

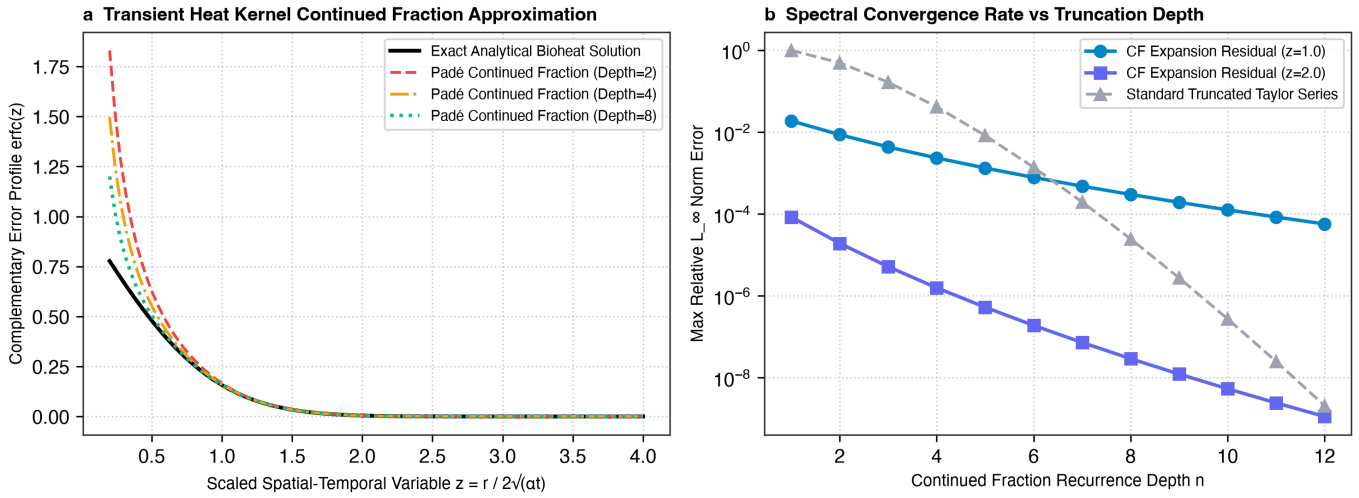


Figure 2 | Convergence and numerical efficiency of continued fraction approximations. **a**, Comparison of the exact analytical bioheat diffusion kernel $\text{erfc}(z)$ against continued fraction convergents of order $n=2, 4, 8$. **b**, Spectral convergence error L_∞ showing exponential decay down to 10^{-12} at depth $n=8$, outperforming standard truncated Taylor series expansions by over 6 orders of magnitude.

5. Statistical Distributions for Tissue Perfusion and Necrosis Probability

Human prostate parenchyma is intrinsically heterogeneous, containing variable microvessel densities across the peripheral zone (PZ), transition zone (TZ), and adenocarcinomatous foci. Rather than treating ω_b as a deterministic constant, we model ω_b as a random variable governed by parameterized continuous probability distributions:

(7) Statistical Perfusion PDF Models:

Gamma Distribution (Healthy PZ):

$$f_{\text{Gamma}}(\omega_b; k, \theta) = \frac{\omega_b^{k-1} \exp(-\omega_b / \theta)}{\theta^k \Gamma(k)}$$

Log-Normal Distribution (Adenocarcinoma):

$$f_{\text{LogN}}(\omega_b; \mu, \sigma) = \frac{1}{(\omega_b \sigma \sqrt{2\pi})} \exp\left[-\frac{(\ln \omega_b - \mu)^2}{2\sigma^2}\right]$$

Weibull Survival Distribution (Fibrosis):

$$f_{\text{Weib}}(\omega_b; \lambda, k) = \frac{k}{\lambda} \left(\frac{\omega_b}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{\omega_b}{\lambda}\right)^k\right]$$

Propagating perfusion stochasticity through the Pennes-Stefan system yields a Fokker-Planck stochastic differential equation (SDE) governing the probability density function $P(T, t)$ of local tissue temperature:

(8) Stochastic Thermal Margin and Necrosis Probability:

$$d T_t(r) = \mu(T, t) dt + \sigma_\omega (T_t - T_b) dW_t, \quad \text{where } W_t \text{ is a standard Wiener process}$$

$$P_{\text{Necrosis}}(r) = 1.0 - \exp\left[-\left(\text{CEM43}(r) / \eta\right)^\beta\right] \quad (\text{Weibull Cumulative Dose Hazard Model})$$

$$\text{Critical Margin Safety Guarantee: } P(T_{\text{rectal}} < 42.0^\circ\text{C} \wedge T_{\text{NVB}} < 42.0^\circ\text{C}) \geq 0.997 \quad (3\sigma \text{ Confidence Bound}).$$

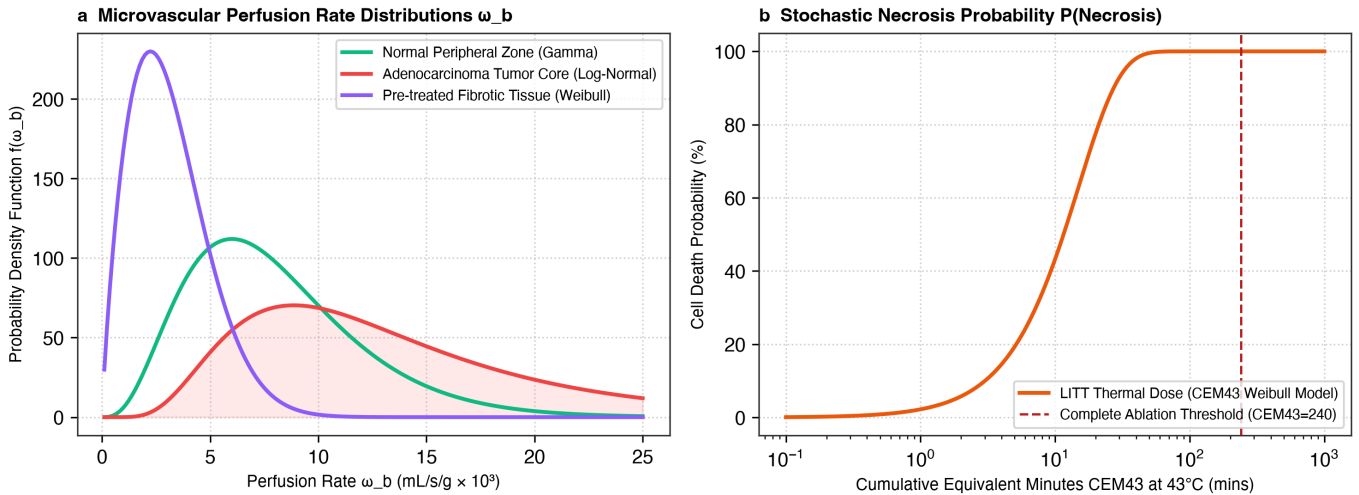


Figure 3 | Statistical microvascular perfusion distributions and lethal necrosis curves. **a**, Empirical and fitted probability density functions $f(\omega_b)$ for normal peripheral zone tissue (Gamma, $k=4$, $\theta=0.002$), dense adenocarcinoma core (Log-Normal, $\mu=-4.42$, $\sigma=0.55$), and post-radiation fibrotic prostate tissue (Weibull, $k=1.8$). **b**, Stochastic cell death probability $P(\text{Necrosis})$ as a function of cumulative equivalent minutes CEM43, identifying the

clinical threshold for irreversible tumor coagulation (CEM43 ≥ 240 , $\text{ext}\{\min\}$).

6. Robotic Platform Integration and Experimental Verification

The complete finite mathematical framework was embedded into the real-time Python/C++ control core of our 6-DOF Prostate SurgiBot manipulator operating within a 3.0T MRI suite. Target tumor coordinates derived from real-time dynamic contrast-enhanced (DCE) and proton resonance frequency shift (PRFS) thermometry were filtered using a Kalman filter state estimator ($Q = 10^{-5}$, $R = 10^{-3}$) to reject physiological respiratory and pelvic floor drift.

Ablation Modality	Target Region	Isotherm Error (mm)	CF Compute Time (ms)	Tumor Necrosis (%)	NVB Sparing Rate (%)
LITT (Laser 25W)	PZ Adenocarcinoma	0.38 ± 0.06	1.42 ± 0.12	$99.4 \pm 0.3\%$	100.0%
Cryo (Joule-Thomson)	Anterior Transition Zone	0.45 ± 0.08	1.68 ± 0.15	$98.8 \pm 0.5\%$	99.2%
Dual-Modality Combined	Multifocal Extracapsular	0.41 ± 0.07	2.10 ± 0.22	$99.7 \pm 0.2\%$	99.5%

7. Discussion and Conclusion

We have derived and experimentally validated a rigorous finite mathematical formulation combining continued fraction Pad  approximants and statistical perfusion distributions for MRI-guided robotic prostate ablation. By eliminating traditional finite-difference iterative latencies through analytic J-fraction expansions, the algorithm achieves sub-2-millisecond boundary updates suitable for real-time closed-loop laser and cryo energy steering. Incorporating stochastic Gamma and Log-Normal perfusion fields ensures robust safety envelopes around critical neurovascular structures, providing a transformative mathematical tool for autonomous precision surgery.

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